Advances: (1) lack of data on witness officers, (2) dynamic programming for Poisson-Multinomial

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1 Conditional ordinal stereotype model still applies in the absence of data on witness officers

If data are available only on officers who use force (no witness officers), then the unconditional likelihood for a single incident is

$$P(Y_1 = y_1, \ldots, Y_m = y_m \mid Y_1 > 0, \ldots, Y_m > 0, \mathbf{z}, \mathbf{h}, \mathbf{s}, \boldsymbol{\lambda}) \tag{1}$$

The conditioning on $Y_i > 0$ carries through to the conditional likelihood.

$$CL_\ell(\boldsymbol{\lambda}) = \frac{\prod_{i=1}^m P(Y_i = y_i \mid Y_i > 0, \mathbf{z}_\ell, \mathbf{h}, \mathbf{s}, \lambda_i)}{\sum_{\mathbf{y}^* \in \mathcal{K}} \prod_{i=1}^m P(Y_i = y_i^* \mid Y_i > 0, \mathbf{z}_\ell, \mathbf{h}, \mathbf{s}, \lambda_i)} \tag{2}$$

Modifying Anderson's ordinal stereotype model to a truncated version that conditions on $Y > 0$ yields

$$P(Y = y|Y > 0) = \frac{e^{h_y(\mathbf{z})+s_y\lambda}}{\sum_{j=0}^J e^{h_j(\mathbf{z})+s_j\lambda}} \frac{\sum_{j=0}^J e^{h_j(\mathbf{z})+s_j\lambda}}{\sum_{j=1}^J e^{h_j(\mathbf{z})+s_j\lambda}} \tag{3}$$

$$= \frac{e^{h_y(\mathbf{z})+s_y\lambda}}{\sum_{j=1}^J e^{h_j(\mathbf{z})+s_j\lambda}}, \quad \text{for } y = 1, \ldots, J \tag{4}$$

where the second term in (3) is $1/P(Y > 0)$.

Equations (5)-(8) in Ridgeway (nd) are essentially unchanged as a result, only the normalizing sums start at $j = 1$ instead of $j = 0$.

$$CL_\ell(\mathbf{h}, \mathbf{s}, \boldsymbol{\lambda} \mid \mathbf{k}) = \frac{\prod_{i=1}^m \frac{\exp\left(h_{y_i}(\mathbf{z})+s_{y_i}\lambda_i\right)}{\sum_{j=1}^J \exp\left(h_j(\mathbf{z})+s_j\lambda_i\right)}}{\sum_{\mathbf{y}^* \in \mathcal{K}} \prod_{i=1}^m \frac{\exp\left(h_{y_i^*}(\mathbf{z})+s_{y_i^*}\lambda_i\right)}{\sum_{j=1}^J \exp\left(h_j(\mathbf{z})+s_j\lambda_i\right)}} \tag{5}$$

$$= \frac{\exp\left(\sum_{i=1}^m s_{y_i}\lambda_i\right)}{\sum_{\mathbf{y}^* \in \mathcal{K}} \exp\left(\sum_{i=1}^m s_{y_i^*}\lambda_i\right)} \tag{6}$$

Since we never observe $y = 0$, s_0 never appears in the conditional likelihood, introducing a new identifiability issue. We need to shift the constraints on $\mathbf{s}$ so that we have a new $s_j^* = \frac{s_j - 1}{s_2 - 1}$ for $j = 1, \ldots, J$.

To estimate the parameters in the absence of data on witness officers, the same algorithm applies. Adjust the use of force outcome to $y^* = y - 1$, shifting all the force types down one level and apply the algorithm. The estimands, however, will be $\mathbf{s}^*$ and $(s_2 - 1)\hat{\boldsymbol{\lambda}}$ instead of the original parameters. Flagging outliers proceeds as before.

2 A dynamic program for computing the denominator

The denominator is computable via a dynamic program that recursively accumulates probability mass over partial count vectors, equivalent to extracting a single coefficient from a product of multinomial generating functions.

Let $y_i \in \{0, \ldots, J\}$ denote the observed force level for officer $i = 1, \ldots, m$, and define the counts

$$k_j = \sum_{i=1}^{m} \mathbf{1}\{y_i = j\}, \quad j = 0, \ldots, J \tag{7}$$

Define the category probabilities

$$p_{ij} = \frac{\exp(s_j \lambda_i)}{\sum_{j^*=0}^{J} \exp(s_{j^*} \lambda_i)} \tag{8}$$

The denominator of an incident's contribution to the conditional likelihood can be written as a product of independent terms and a Poisson–multinomial probability. Let $\mathcal{K}$ be the set of all assignments $\mathbf{y}^* = (y_1^*, \ldots, y_m^*)$ whose category counts equal $\mathbf{k}$.

$$\begin{aligned}
\sum_{\mathbf{y}^* \in \mathcal{K}} \exp\left(\sum_{i=1}^{m} s_{y_i^*} \lambda_i\right) &= \sum_{\mathbf{y}^* \in \mathcal{K}} \prod_{i=1}^{m} \exp\left(s_{y_i^*} \lambda_i\right) && (9)\\
&= \sum_{\mathbf{y}^* \in \mathcal{K}} \left[\prod_{i=1}^{m} \left(\sum_{j=0}^{J} \exp(s_j \lambda_i)\right) p_{i y_i^*}\right] && (10)\\
&= \left(\prod_{i=1}^{m} \sum_{j=0}^{J} \exp(s_j \lambda_i)\right) \sum_{\mathbf{y}^* \in \mathcal{K}} \prod_{i=1}^{m} p_{i y_i^*} && (11)\\
&= \left(\prod_{i=1}^{m} \sum_{j=0}^{J} \exp(s_j \lambda_i)\right) \mathrm{P}(\mathbf{K} = \mathbf{k}) && (12)
\end{aligned}$$

where $\mathrm{P}(\mathbf{K} = \mathbf{k})$ is the Poisson–multinomial probability induced by the p_{ij}. Note that the summation is restricted to configurations $\mathbf{y}^* \in \mathcal{K}$ that yield the fixed count vector $\mathbf{k}$, so the expression evaluates a single Poisson–multinomial probability rather than the full distribution.

A dynamic program over partial count vectors evaluates this quantity efficiently. Let $\mathbf{u} = (u_0, \ldots, u_{J-1})$, where u_j denotes the number of assignments to category j. For $r = 0, \ldots, m$, define

$$F_r(\mathbf{u}) = \mathrm{P}\left(\sum_{i=1}^{r} \mathbf{1}(Y_i = j) = u_j \text{ for } j = 0, \ldots, J-1\right) \tag{13}$$

The count for category J is implicit and given by

$$u_J = r - \sum_{j=0}^{J-1} u_j \tag{14}$$

The recursion is initialized at

$$F_0(\mathbf{u}) = \begin{cases} 1 & \text{if } \mathbf{u} = \mathbf{0} \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

For $r = 1, \ldots, m$, the recursion is

$$F_r(\mathbf{u}) = p_{rJ} F_{r-1}(\mathbf{u}) \; + \; \sum_{j=0}^{J-1} p_{rj} F_{r-1}(\mathbf{u} - \mathbf{e}_j) \tag{16}$$

where $\mathbf{e}_j$ is the jth standard basis vector, and $F_{r-1}(\mathbf{v}) = 0$ for any $\mathbf{v}$ with a negative component. The recursion is evaluated only over feasible states satisfying

$$\max(0, r - k_J) \;\leq\; \sum_{j=0}^{J-1} u_j \;\leq\; r \quad \text{and} \quad 0 \leq u_j \leq k_j \;\text{ for all } j = 0, \ldots, J-1 \tag{17}$$

The desired probability is obtained as

$$\mathrm{P}(\mathbf{K} = \mathbf{k}) = F_m(k_0, \ldots, k_{J-1}) \tag{18}$$

Finally, the log normalizing constant used in the conditional likelihood is

$$\sum_{i=1}^{m} \left(\log \sum_{j=0}^{J} \exp(s_j \lambda_i) \right) + \log \mathrm{P}(\mathbf{K} = \mathbf{k}) \tag{19}$$

For numerical stability, at each iteration r, the values $F_r(\mathbf{u})$ are rescaled by their maximum over $\mathbf{u}$ to mitigate underflow, and the corresponding log-scale factors are accumulated and added back at the end of the recursion.

The time complexity is $O\left(m \prod_{j=0}^{J-1}(k_j + 1)\right)$ for one incident with m officers. In the worst case over $\mathbf{k}$ ($k_J = 0$ and the first J counts are as equal as possible), the time complexity is $O\left(m \left(\frac{m}{J} + 1\right)^J\right)$, which for a fixed J is $O\left(m^{J+1}\right)$.

To examine empirical scaling, I conducted a simulation study varying incident size from $m = 2$ to $m = 20$, with 10 replications at each value of m. For each replication, officer-specific parameters λ_i were drawn independently from $\mathrm{U}(-2, 2)$, and force levels were generated by sampling $y_i \in \{0, 1, 2, 3\}$ from a multinomial model with probabilities proportional to $\exp(s_j \lambda_i)$ for fixed scores $\mathbf{s} = (0, 1, 1.5, 2)$. For each simulated incident, I recorded the quantity $\prod_{j=0}^{J-1}(k_j + 1)$ and measured the average runtime of the dynamic program. To stabilize timing estimates across incident sizes, the number of repetitions was scaled inversely with m, holding the total computational effort approximately constant across simulations.

Empirical timing results shown in Figure 1 indicate substantially weaker scaling than the worst-case bound. In a log-log regression, computation time was $O\left(m^0 \left(\prod_{j=0}^{J-1}(k_j + 1)\right)^{0.35}\right)$, primarily driven by the size of the feasible state space $\prod_{j=0}^{J-1}(k_j + 1)$, with little additional dependence on m. Moreover, the relationship was sublinear, reflecting that many states are infeasible or have zero probability, so the effective number of active states is much smaller than the full grid.

Worked example Consider an incident with $m = 3$ officers and $J = 3$, so that the force levels are $\{0, 1, 2, 3\}$. Suppose the observed force vector is

$$\mathbf{y} = (y_1, y_2, y_3) = (0, 1, 1) \tag{20}$$

so that the observed counts are

$$\mathbf{k} = (k_0, k_1, k_2, k_3) = (1, 2, 0, 0) \tag{21}$$

Then

$$\mathcal{K} = \{(0, 1, 1), (1, 0, 1), (1, 1, 0)\} \tag{22}$$

the set of all assignments with one officer at level 0, two officers at level 1, and no officers at levels 2 or 3. The denominator is therefore

$$\sum_{\mathbf{y}^* \in \mathcal{K}} \exp\left(\sum_{i=1}^{3} s_{y_i^*} \lambda_i\right) = \exp(s_0\lambda_1 + s_1\lambda_2 + s_1\lambda_3) \tag{23}$$

$$+ \exp(s_1\lambda_1 + s_0\lambda_2 + s_1\lambda_3) \tag{24}$$

$$+ \exp(s_1\lambda_1 + s_1\lambda_2 + s_0\lambda_3) \tag{25}$$

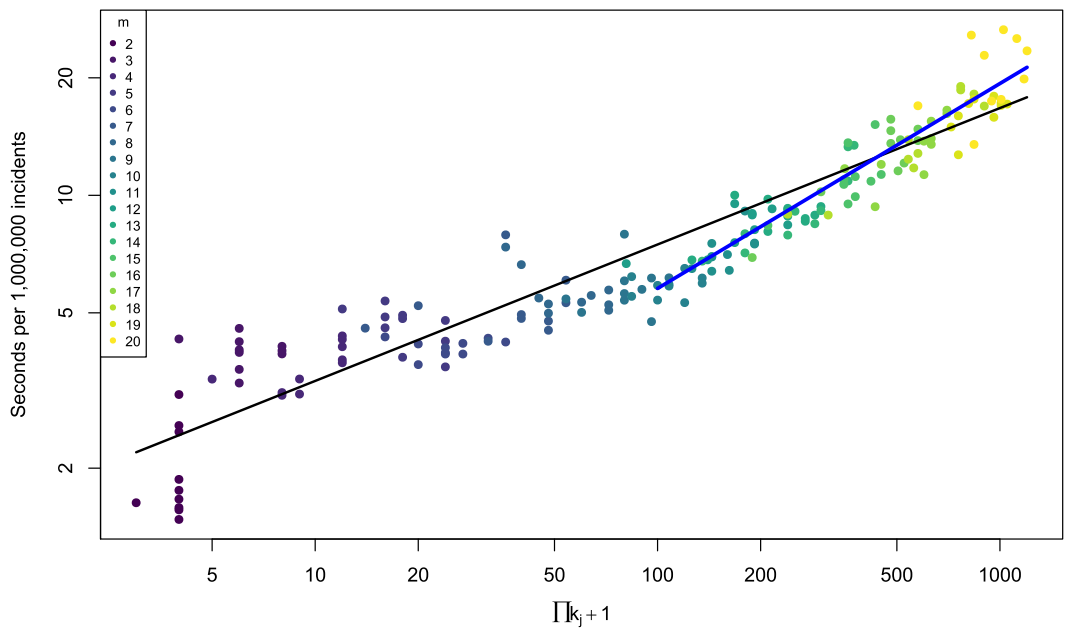

Figure 1: Empirical time complexity of dynamic program. $\prod k_j + 1$ primarily drives computation time. Slope over 3 to 1200 (0.35) and slope over 100 to 1200 (0.52) indicate sublinear complexity empirically

Let

$$p_{ij} = \frac{\exp(s_j \lambda_i)}{\sum_{j^*=0}^{3} \exp(s_{j^*} \lambda_i)} \tag{26}$$

Then the corresponding Poisson–multinomial probability is

$$\mathrm{P}(\mathbf{K} = \mathbf{k}) = p_{10}p_{21}p_{31} + p_{11}p_{20}p_{31} + p_{11}p_{21}p_{30} \tag{27}$$

The aim of this example is to show that the dynamic program yields the same result.

The dynamic program tracks partial counts

$$\mathbf{u} = (u_0, u_1, u_2) \tag{28}$$

with the count for category 3 given implicitly by

$$u_3 = r - u_0 - u_1 - u_2 \tag{29}$$

The target state is

$$(k_0, k_1, k_2) = (1, 2, 0) \tag{30}$$

The recursion is

$$F_r(u_0, u_1, u_2) = p_{r3}F_{r-1}(u_0, u_1, u_2) + p_{r0}F_{r-1}(u_0 - 1, u_1, u_2) \tag{31}$$

$$+ p_{r1}F_{r-1}(u_0, u_1 - 1, u_2) + p_{r2}F_{r-1}(u_0, u_1, u_2 - 1) \tag{32}$$

with $F_0(0, 0, 0) = 1$ and all other initial values equal to zero.

After one officer, $r = 1$ and the feasible states satisfy

$$1 - k_3 \le u_0 + u_1 + u_2 \le 1, \quad 0 \le u_j \le k_j \tag{33}$$

Since $k_3 = 0$ and $k_2 = 0$, the only relevant states are

$$F_1(1,0,0) = p_{13}F_0(1,0,0) + p_{10}F_0(0,0,0) + p_{11}F_0(1,-1,0) + p_{12}F_0(1,0,-1) \tag{34}$$
$$= p_{10} \tag{35}$$
$$F_1(0,1,0) = p_{13}F_0(0,1,0) + p_{10}F_0(-1,1,0) + p_{11}F_0(0,0,0) + p_{12}F_0(0,1,-1) \tag{36}$$
$$= p_{11} \tag{37}$$

After two officers, $r = 2$ and

$$F_2(1,1,0) = p_{23}F_1(1,1,0) + p_{20}F_1(0,1,0) + p_{21}F_1(1,0,0) + p_{22}F_1(1,1,-1) \tag{38}$$
$$= p_{20}p_{11} + p_{21}p_{10} \tag{39}$$
$$F_2(0,2,0) = p_{23}F_1(0,2,0) + p_{20}F_1(-1,2,0) + p_{21}F_1(0,1,0) + p_{22}F_1(0,2,-1) \tag{40}$$
$$= p_{21}p_{11} \tag{41}$$

To compute the target state $F_3(1,2,0)$, these are the only predecessor states that need to be evaluated explicitly.

After three officers, $r = 3$ we compute

$$F_3(1,2,0) = p_{33}F_2(1,2,0) + p_{30}F_2(0,2,0) + p_{31}F_2(1,1,0) + p_{32}F_2(1,2,-1) \tag{42}$$
$$= p_{30}p_{21}p_{11} + p_{31}(p_{20}p_{11} + p_{21}p_{10}) \tag{43}$$
$$= p_{10}p_{21}p_{31} + p_{11}p_{20}p_{31} + p_{11}p_{21}p_{30} \tag{44}$$

which equals the result from the complete enumeration in (27). Thus

$$F_3(1,2,0) = \mathrm{P}(\mathbf{K} = (1,2,0,0)) \tag{45}$$

showing that the dynamic program computes the single Poisson–multinomial probability associated with the observed count vector.

References

Ridgeway, G. (n.d.). A conditional ordinal stereotype model to estimate police officers' propensity to escalate force. *Journal of the American Statistical Association*, 1–20.